SΔϕ-09 — Error, Boundary, and Freedom

as Re-entry Dynamics in the ARF Closure
within the Sofience–Δϕ Formalism

Series	SΔϕ (The Sofience–Δϕ Formalism)
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Abstract

This document formalizes three derived notions—Error, Boundary, and Freedom—within the ARF closure of the Sofience–Δϕ Formalism. Error is defined as an irreversible transition residue that fails to re-enter the system's recursive update loop, thereby exposing structural limits. Boundary is defined as the historically accumulated marking of such error events, constraining subsequent transitions. Freedom is redefined as sustained re-opening: the persistence of path-difference re-entry despite boundary formation. Together, these axioms locate "error" not as moral wrongness but as a limit-exposure event that generates topology.

0. Position in Series

This document extends SΔϕ-08 (ARF Closure) by defining three derived notions:
• Error as non-reintegrable transition residue,
• Boundary as accumulated error-mark topology,
• Freedom as sustained re-opening under boundary formation.

1. Minimal primitives

Assume the primitives established in the series:
• Δϕ: irreversible phase-change / transition residue
• A (Agency): recursive transition-law update
• R (Responsibility): irreversible cost internalization
• F (Freedom): persistent re-entry of path-difference into subsequent transitions

Let the system state be ϕ_t. Let the transition operator be G, and the minimal marker mechanism be H.

2. Core definitions

D1 — Re-entry

A transition residue $\Delta\phi_t$ is said to re-enter iff it is reintroduced as an effective constraint/input into the next transition:
$\Delta\phi_t \to G(\phi_t, \Delta\phi_t) \to \phi_{t+1}$
Operationally: the system's next-state generation depends on the prior residue in a non-trivial way.

D2 — Error (Non-reintegrable transition)

An Error occurs when an irreversible transition residue is produced, but fails to re-enter the system's recursive loop as a constraint/input:
$\Delta\phi_t \neq 0 \wedge \neg\text{re-entry}(\Delta\phi_t)$
Interpretation: "a real transition happened, but the system cannot incorporate it into its own history."

D3 — Boundary (Accumulated error mark)

A Boundary is the stabilized/accumulated effect of error events, represented as a persistent mark B that restricts future transitions:
$B_{t+1} = B_t \oplus M(\text{Error}_t)$
Boundary is not pre-given; it is generated historically.

D4 — Freedom (Sustained re-opening under boundary)

Within ARF, Freedom is not the number of choices but the capacity for path-difference to persist as re-entry, even while boundaries form:
$F \Leftrightarrow \text{re-entry}(\Delta\phi_t)$ persists across t despite growth of B_t
Equivalently: the system does not collapse into closure solely due to accumulated constraints.

3. Minimal axioms

Axiom 1 — Irreversibility

$\Delta\phi_t \neq 0 \Rightarrow$ the transition cannot be undone.

Axiom 2 — ARF Closure baseline (inherited from SΔϕ-08)

A stable operational loop requires the co-presence of: A (the system can update transition law), R (update induces irreversible internalized cost), and F (path-difference re-enters recursively).

Axiom 3 — Error as re-entry failure

$\text{Error}_t \Leftrightarrow (\Delta\phi_t \neq 0) \wedge \neg\text{re-entry}(\Delta\phi_t)$.

Axiom 4 — Boundary as error accumulation

$B_{t+1} = \text{Update}(B_t, \text{Error}_t)$. Boundary is generated only through encountered failure.

Axiom 5 — Freedom as non-collapse

Freedom holds when the system maintains re-entry capacity despite boundary growth.

4. Derived propositions

P1 — Error is non-moral

Error is not “wrongness” but a limit exposure event: Error ⇒ limit becomes observable.

P2 — Boundary is historical

If no error occurs, no boundary is generated: $(\forall t, \neg \mathrm{Error}_t) \Rightarrow B_t = B_0$.

P3 — Freedom generates boundary indirectly

Greater exploratory re-entry increases boundary formation probability: F ↑ ⇒ Pr(Error) ↑ ⇒ B ↑, yet Freedom is defined by survival of re-opening, not by minimizing errors.

5. Collapse conditions

Three collapse modes mirror ARF logic:
• Drift: A without R/F stabilizing re-entry
• Fixation: R without A/F (over-constraint)
• Illusion: F-like branching without R-integrated cost (non-grounded divergence)

6. Series references (DOIs)

• SΔϕ-01 — Sofience–Δϕ Minimal Core: Phase-Change Formalism of Existence (v1.1)

• SΔϕ-02 — Subject as Interpretive Emergence: Extension of the Sofience–Δϕ Minimal Core (v1.1)

• SΔϕ-03 — Irreversibility as the Minimal Condition of Existence (v1.1)

• SΔϕ-07 — Hmin Interpretation: Minimal Axioms (v1.0)

• SΔϕ-08 — Agency–Responsibility–Freedom Closure: Minimal Axiomatic Formulation (v1.0)